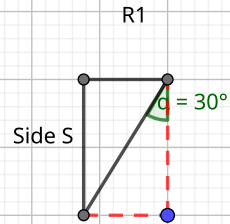


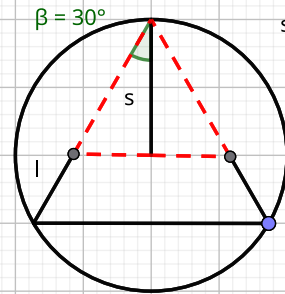
Non regular polygon (rectangle R1)



triangle is T1

the red triangle, T2

circle is C1



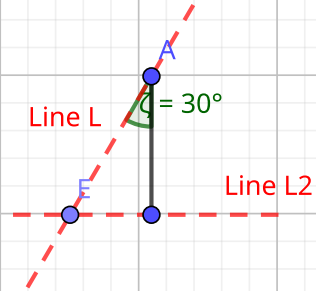
s, equals radius of C1, which is two

Since α is the same as β , we can create the rectangle R1

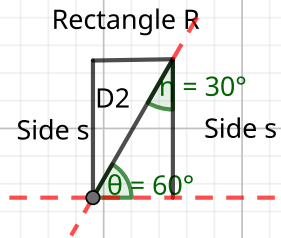
the triangle, which sides are red, is inside the circle at the right (that is denoted in black)

height of triangle T2, is equal to radius of circle. $r = 2$ (radius of circle)

Since, we have angle (30°) , we can construct the rectangle R1

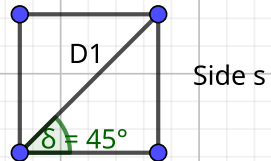


From above, we can construct R



Now we create a square, to handle problem (see below calculus)

on square and rectangle, side s is equal



Formula for getting a square diagonal

$$D1 = s * \sqrt{2}$$

s, above is side of square, and side of rectangle

calculus

$$D1 = 2 * \sqrt{2}$$

using sin

$$\sin(45) = \frac{s}{D1} \quad (1)$$

$$\sin(60) = \frac{s}{D2} \quad (2)$$

multiplication D1 by both sides on (1)

$$D1 * \sin(45) = s$$

multiplication by D2 on (2)

$$D2 * \sin(60) = s$$

$$D1 * \sin(45) = D2 * \sin(60)$$

divide both sides by $\sin(60)$

$$D1 * \frac{\sin(45)}{\sin(60)} = D2$$

divide both sides by D1

$$\frac{\sin(45)}{\sin(60)} = \frac{D2}{D1}$$

We have

$$\sin(45) = 0.707106781$$

$$\sin(60) = 0.866025404$$

Assigning Diagonal of square to D1, and convert sins to values.

$$\frac{0.707106781}{0.866025405} = \frac{D2}{2 * \sqrt{2}}$$

multiply both sides by $2 * \sqrt{2}$

$$\frac{2}{0.866025405} = D2$$

$$D2 = 2.309401076$$

Observation

if i multiply $\cos(45)$ by 4, i get the diagonal of square. with side 2

$$\cos(45) = \frac{\sqrt{2}}{2}$$

$$4 * \cos(45) = 4 * \frac{\sqrt{2}}{2}$$